Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/23/2024
PI Name:	Dai, Guohao Ph.D.		
Project Title:	Bioengineer Long-lasting 3D Neurovascular Microphysiological System to Model Chronic Inflammation-mediated Neurodegeneration		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	NIH, BARDA, FDA	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	g.dai@northeastern.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-373-2207
Organization Name:	Northeastern University		
PI Address 1:	360 Huntington Ave.		
PI Address 2:	ISEC 224		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02115-5005	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA015N. Extended Longevity of 3D Tissues and Microphysiological Systems for Modeling of Acute and Chronic Exposures to Stressors
Start Date:	02/21/2023	**End Date:**	02/14/2027
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
Contact Monitor:	Koniges, Ursula	**Contact Phone:**	202-256-8786
Contact Email:	ursula.m.koniges@nasa.gov		
Flight Program:			
Flight Assignment:	Per the Research Announcement for this solicitation: this effort is a collaboration between NASA; the National Institutes of Health (NIH); the Biomedical Advanced Research and Development Authority (BARDA); and the Food and Drug Administration (FDA).		
Key Personnel Changes/Previous PI:	No change		
COI Name (Institution):	Friedel, Roland Ph.D. (ICAHN School Of Medicine at Mount Sinai) Koppes, Abigail Ph.D. (Northeastern University) Koppes, Ryan Ph.D. (Northeastern University) Zou, Hongyan Ph.D., M.D. (ICAHN School Of Medicine at Mount Sinai)		
Grant/Contract No.:	80ARC023CA005		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	One of the most significant obstacles in building a long-lasting 3D tissue model is the lack of a stable functional vasculature. The blood vessels not only transport nutrients and oxygen, but also secrete important angiocrine factors to instruct organ/tissue homeostasis. The continuous regression of vasculature in current models will disrupt perfusion and lead to tissue necrosis. To tackle this challenge, various techniques have been attempted, such as improving biomaterial hydrogels and perfusion systems. These efforts are valuable to prolong the vascular life span, but only to a limited extent, and 6 months remains a far-reaching goal. Two fundamental issues preclude establishing a long-lasting vasculature in vitro: 1) intrinsic: cultured human endothelial cells (ECs) gradually lose their in vivo functions and tissue adaptability; 2) extrinsic: long-term maintenance of ECs requires serum/growth cytokines, resulting in loss of their innate angiogenic properties. Solving these fundamental issues is critical for building a stable long-lasting vasculature in vitro. Our recent advances have led to the central hypothesis that reprogramming ECs by forced expression of key factors can restore their in vivo characteristics, such as durable lumens, less dependence on serum/growth cytokines, and hemodynamic/tissue adaptability. To study neurovascular interactions, we have developed a 3D human brain vasculature system that consists of human brain ECs, pericytes, and astrocytes. This system has interconnected open lumens and tight junctions with permeability similar to the blood brain barrier. Building on this platform, we developed a biomimetic 3D neurovascular model that can support the homeostatic balance of human neural stem cell (NSC) derived from induced pluripotent stem cells (iPSCs) for a 1 month period, including self-renewal, neuronal maturation, and NSC quiescence. The goal of this project is to extend the lifespan of existing 3D neurovascular model from 1 to 6 months and demonstrate its utility by exposing to chronic stressors. In phase one, we will integrate the following approaches to prolong the lifespan of our current model: 1) Reprogram with key factors to rejuvenate in vivo EC functionalities and promote durable vascular lumen; 2) Optimize a serum free medium by screening small molecules critical for the stable vascular lumen; 3) Integrate technologies for prolonged culture and real-time sensing; these include fiber photometry that allows real-time recording of neuronal activity, fluorescent reporters that allow live visualization of vascular morphology, fluid flow, Ca2+ signaling, oxygen and cellular quiescence, a stable perfusion system with fluid/pressure control for prolonged culture. Combining these approaches, we will validate the stability and functionality of the neurovascular system in terms of vascular morphology, lumen interconnectivity, perfusion, permeability, and neural activity, and NSC reserves for 6 months. In phase two, we will validate the system to model chronic inflammation-mediated neurodegeneration by introducing microglia and inflammatory cytokines and validating whether this approach can faithfully replicate the immune response, neurodegeneration, and premature depletion of NSC reserves over a 6 month period. Significance: Because a stable vasculature is critical for normal tissue function and homeostasis, the success of this project will have broad impact on building many other tissue models for prolonged culture using human cells (both male and female) – heart, kidney, muscle, cancer – to model human diseases.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The success of this project will have broad impact on building many other vascular tissue models for prolonged culture – heart, kidney, muscle, cancer – to model human diseases.
Task Progress:	Specific Aim 1: Reprogram with key factors to rejuvenate in vivo endothelial cell (EC) functionalities. Progress: we have made and validated ETV2 lentivirus, and demonstrated that ETV2-ECs blood vessels last longer than control. We have made and validated E4ORF1 lentivirus, and showed that E4ORF1 made ECs survive longer under low serum condition, and E4ORF1-ECs blood vessels stay longer than control. Specific Aim 2: Optimize culture medium by screening molecules critical for the stable vascular lumen. Progress: we have tested several medium conditions, and found that platelet lysate and S1P can prolong the stability of the bioengineered vessels. Specific Aim 3: Integrate technologies for prolonged culture and real-time sensing. Progress: we have derived neural progenitor cells (NPCs) from induced pluripotent stem cells (iPSCs), and demonstrated that NPCs can be differentiated into neurons in vascular co-culture. To provide real-time recording of neuronal activity, we have made jGCaMP8 lentivirus for real-time Ca2+ sensing, and showed that jGCaMP8 reports the real-time calcium activity in iPSCs derived neurons. We have also designed and fabricated a microfluidic device with successful gel loading with phase guide working. We have integrated a mini-pump to drive interstitial flow (~ul/min); the desired interstitial flow is achieved across hydrogel. However, during the testing of the microfluidic device for culturing cells, we found extensive death of the cells. Troubleshooting experiments pointed to the gas impermeable issue of the materials used for microfluidic device construction. We therefore redesigned and constructed the microfluidic device using polydimethylsiloxane (PDMS) on 3D-printed molds. The experiment is ongoing.
Bibliography Type:	Description: (Last Updated: 04/23/2026)
Articles in Peer-reviewed Journals	Winkelman MA, Dai G. "Bioengineered perfused human brain microvascular networks enhance neuralprogenitor cell survival, neurogenesis, and maturation." Sci Adv. 2023 May 10;9(19):eaaz9499. http://dx.doi.org/10.1126/sciadv.aaz9499 ; PMID: 37163593; PMCID: PMC10171804 , May-2023